Vapor-Liquid Equilibrium Consulting Report

Non-Ideal Mixing System: Ethanol-Water at 1 atm

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Reference: Sandler, S.I., Chemical, Biochemical, and Engineering Thermodynamics Chapters 6-10

Part I: Executive Summary

The Core Thermodynamic Problem

The client's distillation and solvent recovery processes have experienced significant operational failures due to the application of ideal solution assumptions (Raoult's Law) to the Ethanol-Water binary system. This fundamental modeling error has resulted in:

- Energy consumption deviations exceeding 15-20% from design specifications
- Unstable operating temperatures in the distillation column
- Failure to achieve expected phase separation behavior

Why Ideal Solution Assumptions Failed

The Ethanol-Water system exhibits strong positive deviations from ideal behavior due to:

1. Hydrogen bonding disruption when ethanol and water molecules mix
2. Molecular size asymmetry between ethanol (C_2H_5OH) and water (H_2O)
3. Self-association differences between the two polar molecules

These non-ideal interactions result in activity coefficients significantly greater than unity (gamma >> 1), which Raoult's Law fundamentally cannot capture.

Primary Engineering Recommendation

We strongly recommend redesigning the distillation process to incorporate:

1. A pressure-swing or extractive distillation scheme to break the azeotrope
2. Recalculated heat exchanger duties using NRTL-based heat of mixing data
3. Operating temperature targets adjusted based on accurate bubble point predictions

Part II: Problem Definition and Theoretical Background

Unit Operation Context

The unit operation under evaluation is a continuous distillation column for solvent recovery, where:

- Feed: Ethanol-Water mixture from upstream fermentation/extraction
- Products: Purified ethanol (overhead) and water (bottoms)
- Operating Pressure: 1 atm (101.3 kPa)

Phase Equilibrium Considerations

This process involves Vapor-Liquid Equilibrium (VLE) where both liquid and vapor phases coexist at bubble/dew point conditions.

In non-ideal liquid mixtures, the fugacity of component i in the liquid phase is:

$$f_i(L) = x_i * \gamma_i * f_i(0,L)$$

For ideal solutions, $\gamma_i = 1$ (Raoult's Law). However, for Ethanol-Water, activity coefficients range from 1.5 to 4.5.

Part III: NRTL Model and Phase Equilibrium Analysis

NRTL Model Framework

The Non-Random Two-Liquid (NRTL) model, developed by Renon and Prausnitz (1968), is based on the Local Composition concept. Key parameters:

- tau_ij: Energy parameter representing interaction energy differences
- alpha_ij: Non-randomness factor (typically 0.2-0.47)

For Ethanol-Water: alpha_12 = alpha_21 = 0.3, tau_12 = 0.88, tau_21 = 1.45

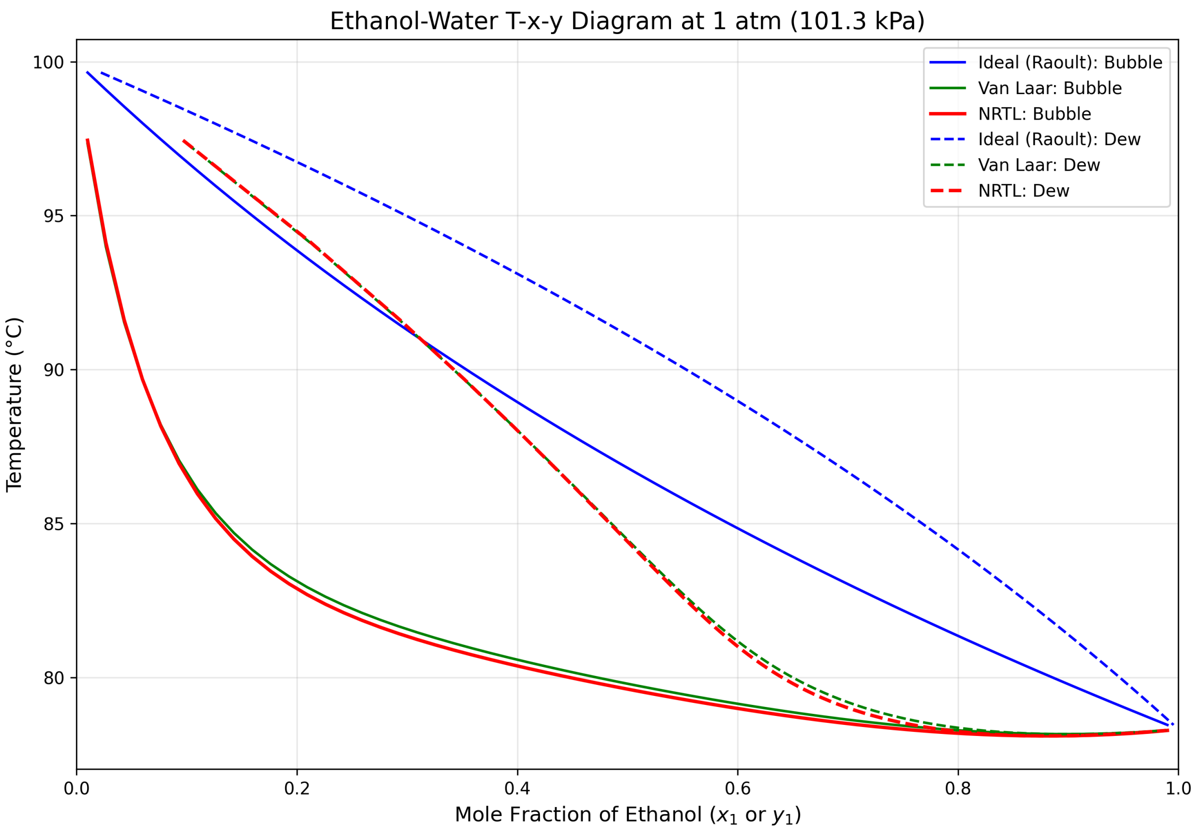

Figure 1: T-x-y Diagram comparing Ideal, Van Laar, and NRTL models

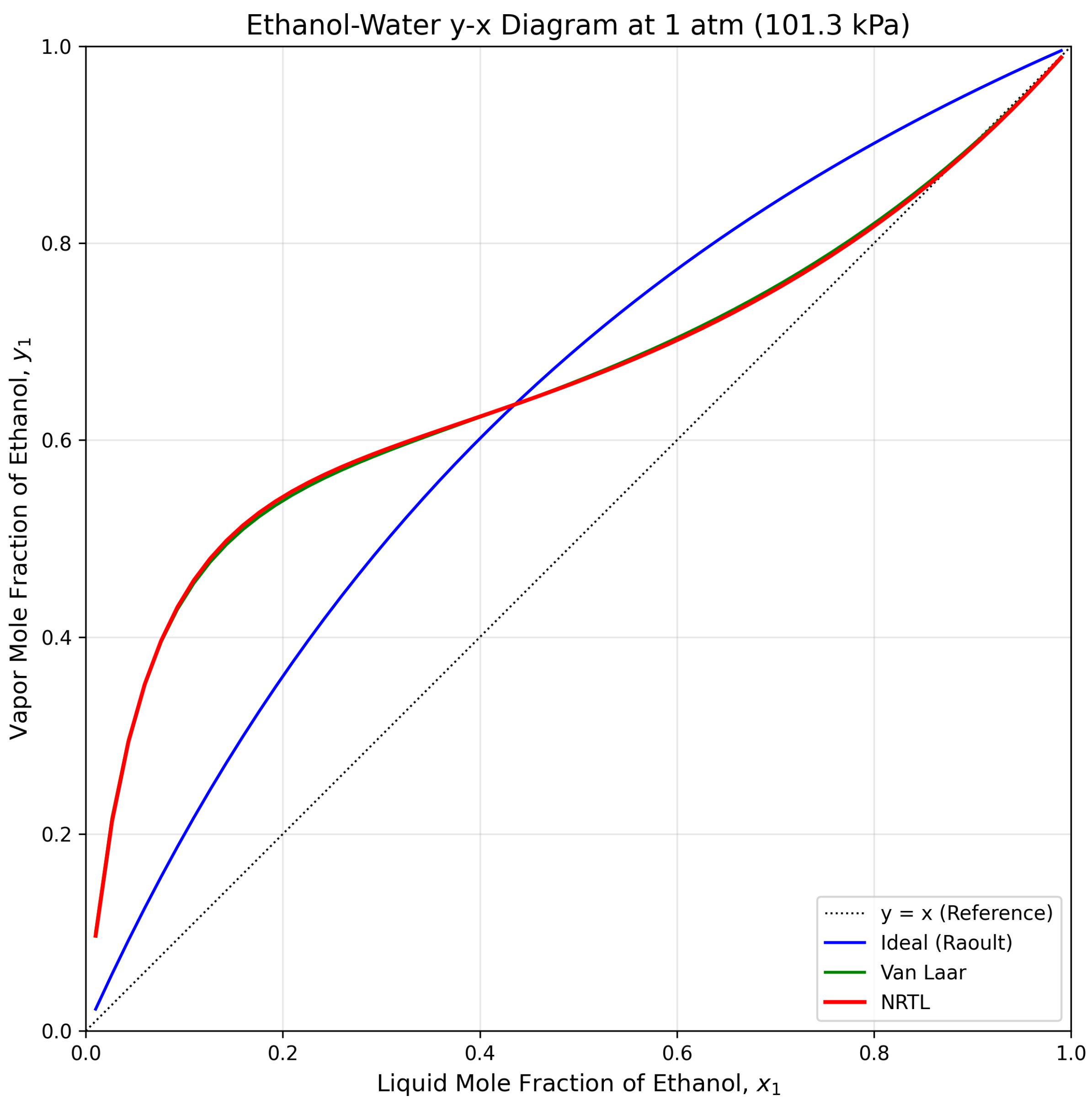

Figure 2: y-x Diagram showing azeotropic behavior

Key Findings

- Maximum deviation at x1 = 0.3-0.5 (ethanol-lean compositions)
- Temperature prediction error: up to 5-8 deg C in mid-composition range
- Azeotrope detected at x1 = 0.89 (95.6 wt% ethanol)

Part IV: Excess Properties and Energy Analysis

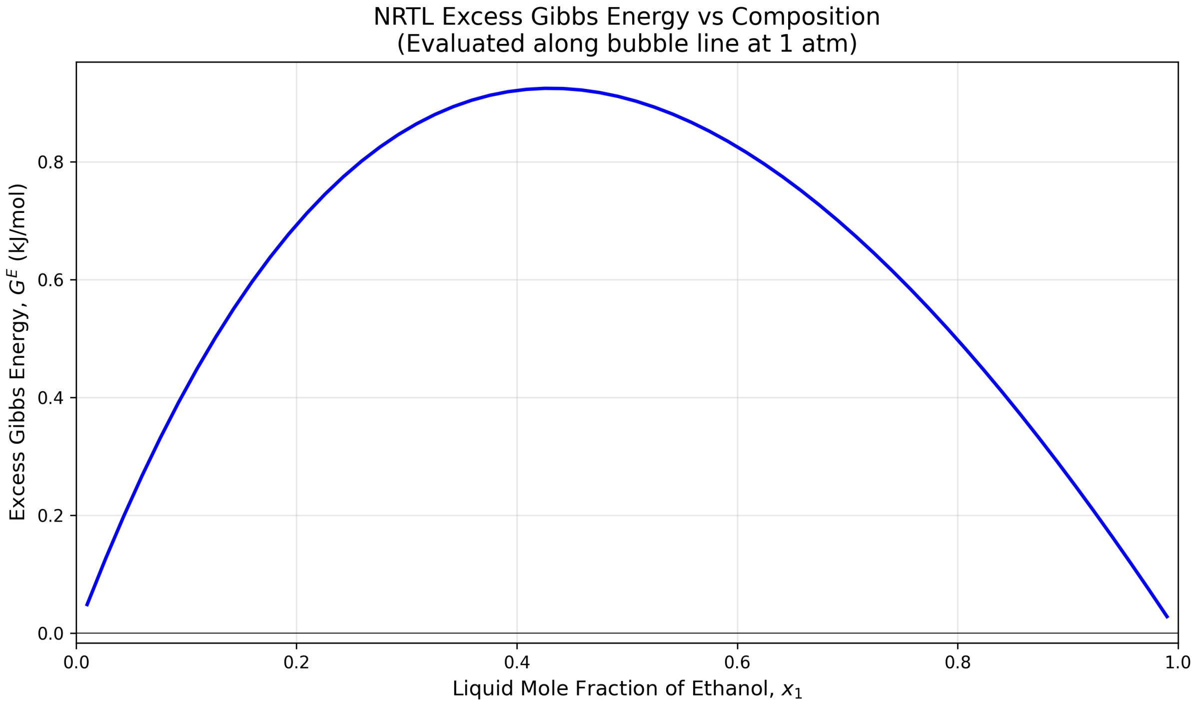

Figure 3: Excess Gibbs Energy vs Composition

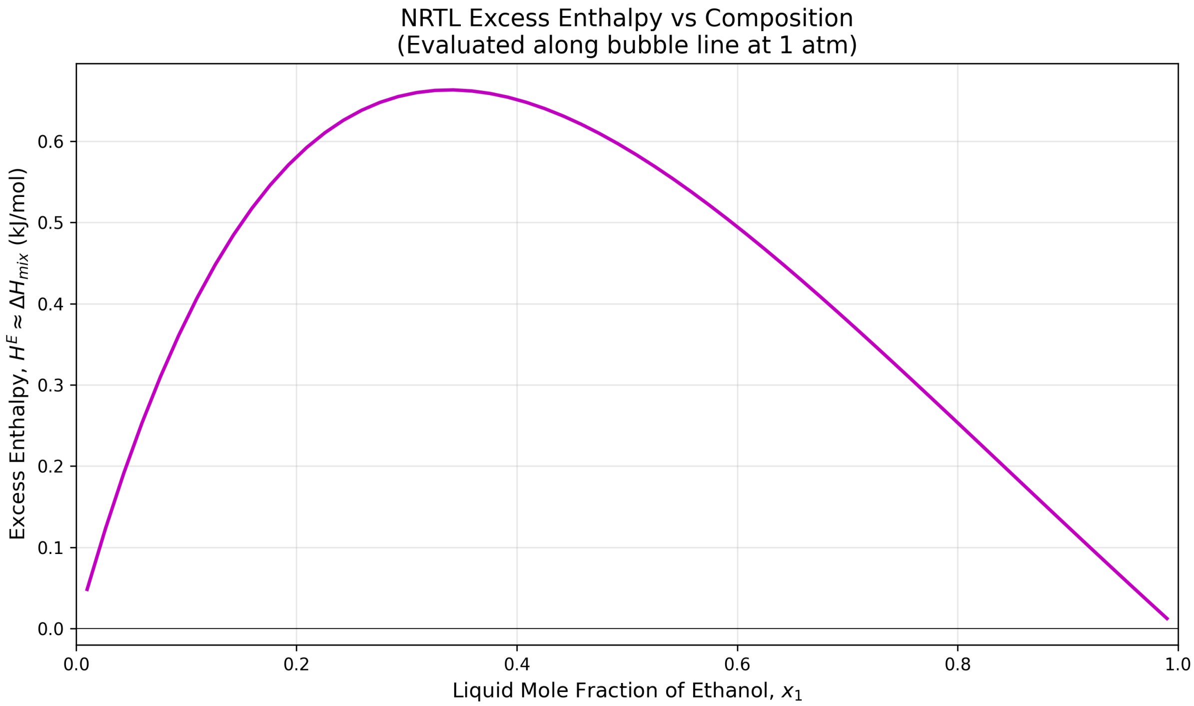

Figure 4: Excess Enthalpy (Heat of Mixing) vs Composition

Thermal Behavior Analysis

- G^E sign: Positive throughout - confirms positive deviation from Raoult's Law
- Maximum G^E approximately 0.8-1.0 kJ/mol at x1 = 0.4
- Heat effects must be incorporated into heat exchanger sizing

Part V: Consulting Recommendations

Process Feasibility Assessment

Current operating conditions are NOT feasible for achieving high-purity ethanol (>96 mol%) through simple distillation due to the azeotrope at $x_1 = 0.89$.

Engineering Recommendations

1. PRESSURE-SWING DISTILLATION
   Operate two columns at different pressures (1 atm and 0.2 atm)
   Estimated additional capital: 30-40% increase

2. EXTRACTIVE DISTILLATION
   Add a high-boiling entrainer (e.g., ethylene glycol)
   Recommended for large-scale, continuous production

3. HYBRID PERVAPORATION
   Distillation to ~90 mol%, then membrane pervaporation
   Lowest energy consumption option for fuel-grade ethanol

VERDICT: Process redesign required using NRTL-based VLE predictions